Predicting Wine Hue

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Summary

We sought to predict wine hue, a variable related to tints of red within the color of wine, based on other chemical and color variables collected from 178 samples of wine across three different cultivars (varieties of plant) in the same Italian region. We compared an initial first order model with these five variables to several second-order and reduced second-order models, concluding in all cases that the first-order model was a better model. We decided to test if dropping the Nonflavanoid_phenols term in the first-order model would still yield an effective model, because this was the only term which did not contribute significant information, and this reduced first-order model was more parsimonious and just as predictive as the previous first-order model. We then checked assumptions of equal variance and normal distribution of the residuals, and found the residuals are homoscedastic and normally distributed.

Original Data (first five observations)

CLASS	ALCOHOL	MALIC_ACID	ASH	ALCALINITY_OF_ASH	MAGNESIUM	TOTAL_PHENOLS	FLAVANOIDS	NONFLAVANOID_PHENOLS	PROANTHOCYANINS	COLOR_INTENSITY	HUE	OD280_OD315_OF_DILUTED_WINES	PROLINE
CLASS 1	14.23	1.71	2.43	15.6	127	2.8	3.06	0.28	2.29	5.64	1.04	3.92	1065
CLASS 1	13.2	1.78	2.14	11.2	100	2.65	2.76	0.26	1.28	4.38	1.05	3.4	1050
CLASS 1	13.16	2.36	2.67	18.6	101	2.8	3.24	0.3	2.81	5.68	1.03	3.17	1185
CLASS 1	14.37	1.95	2.5	16.8	113	3.85	3.49	0.24	2.18	7.8	0.86	3.45	1480
CLASS 1	13.24	2.59	2.87	21	118	2.8	2.69	0.39	1.82	4.32	1.04	2.93	735

Selected Predictor Variables for Hue (from initial exploration of backwards stepwise)

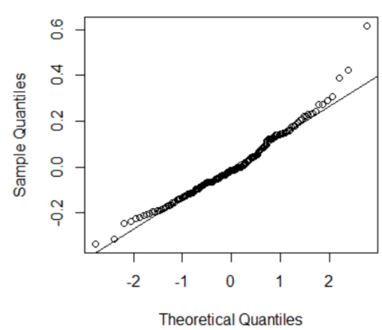
Class = Qualitative variable describing 3 different cultivars of wine

Malic Acid = acid that affects the taste of wine

Nonflavanoid Phenols = Compounds localized in pulp of grapes

Color Intensity = opaqueness of wine

Proline = affects the grape flavor in wine



Final Model

$\hat{Y} = 1.139 + 0.01699x1 - 0.1824x2 - 0.04975x3 - 0.02837x4 + 0.0001608x5$

Where

X1 = 1 if Class = 2, 0 otherwise

X2 = 1 if Class = 3, 0 otherwise

X3 = Malic Acid

X4 = Color Intensity

X5 = Proline

Notable Findings

- A simpler first order model was found to be best due to being the most parsimonious and close to equally predictive as several different second-order models and a slightly more complex first order model
- Although stepwise regression suggested Nonflavanoid Phenols should be considered, this variable was found not to contribute significant information to the model and was excluded from the final model
- Hue and Color Intensity are both color-related variables, yet are not collinear